

# Peer skill identification and social class: Evidence from a referral field experiment\*

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January 31, 2025

## Abstract

We study how accurately peers identify productive others across different skill dimensions. In a lab-in-the-field experiment with 849 university students, we collect incentivized measures of cognitive and social skills and ask peers to refer productive classmates after sustained interaction during the term. We incentivize referrals based on the nominee's measured productivity in skills and randomly assign half the participants to receive additional incentives for identifying high-skilled peers from lower social classes to examine potential social class barriers. We find that peers can successfully identify cognitive skills but not social skills of their classmates. Peers also use academic performance as a proxy for both skills. There is limited evidence of a bias against lower social class peers, and the treatment incentives helps mitigate it. These findings suggest that the accuracy of peer assessments varies substantially across different skill dimensions and

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\*We obtained Institutional Review Board approvals from NYU Abu Dhabi (HRPP 2024-50) and the University of Luxembourg (ERP 24-028). The study design was preregistered in the OSF Registries prior to data collection (see <https://doi.org/10.17605/OSF.IO/V9T3W>).

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appropriate changes in the incentivization structure can make peer assessments robust to existing biases.

**JEL Classification:** C93, D03, D83, J24

**Keywords:** productivity beliefs, referrals, field experiment, skill identification, social class

## 1 Introduction

Evaluating the productivity of others is a standard feature of labor markets. Employers assess job candidates, managers evaluate workers for promotion, and team leaders select collaborators based on beliefs about others' capacity to perform well in different tasks. Whenever observable productivity signals such as test scores or past experience are available, decision-makers rely on those to make accurate evaluations. But such signals are often scarce. In these settings, peer assessments akin to referrals can be a particularly strong screening tool which combines cost-efficiency and accuracy, as sustained interactions among people who work together provide opportunities to directly observe each other's productive qualities in various domains.

However, identifying productive peers across a multitude of productivity dimensions is not straightforward. First, peers could accurately assess productivity in one type of skill, but they may struggle to evaluate it in another. Cognitive and social (interpersonal) skills are two such dimensions of human capital that are increasingly rewarded in labor markets (Deming, 2017, 2023). Second, biases in productivity beliefs can lead to systematic deviations in assessment accuracy. The case for productive individuals from lower social classes is particularly concerning, as they tend to face worse labor market outcomes even when possessing the necessary skills (Stansbury & Rodriguez, 2024).

The overall purpose of this paper is twofold: To evaluate how accurately peers identify productive others in cognitive and social skills, and whether disadvantaged individuals from lower social classes face barriers in selection when peers assess productivity across these skills.

We conducted a lab-in-the-field experiment in a private Colombian university to answer these questions. After interacting for an entire term (about 4 months) in small classrooms (average 26 students per class), participating students assessed classmates' productivity by making referrals. The classroom setting assured that students shared the same referral candidates -their classmates, and we could observe all possible referral choices. As a first step, we collected incentivized cognitive and social skill measures, mapping the productivity distributions of all participants. The advantage of collecting productivity measures for all participants is to properly evaluate the counterfactual and comparing referred peers to those who were not. We then asked for referrals of productive peers. Participants made three separate referrals for each skill and their referral choices could overlap across skills. We incentivized referrals by bonuses contingent on the nominee's productivity in the skill and we did not require referred peers to take any action. Both features allowed us to rule out concerns of potential social transfers (i.e., nepotism or favoritism) or reputational costs typical in the referral literature (see for example [Bandiera, Barankay, and Rasul \(2009\)](#); [Beaman and Magruder \(2012\)](#); [Witte \(2021\)](#)). Once we abstracted away from these elements, the referral decision became one of measuring productivity beliefs through nominated candidates.

Yet, even in an incentivized setting like ours, biases could still be at play because of the underlying beliefs classmates hold about the productivity of their lower social class peers. To address this we designed two treatments. In the baseline treatment, we gave pure performance incentives to referrals regardless of social class. Participants in the quota treatment received additional incentives to identify high-skilled peers from lower social classes. To be able to make comparisons within the same referral choice sets, we assigned half of the participants within each classroom to either treatment. This setup allows us to assess how well incentives can mitigate the said biases in peer productivity beliefs.

Our first goal is understanding how well peers identify cognitive and social skills of their classmates under pure performance incentives at baseline. We find that peers have distinct screening abilities for skills, and use differing referral strategies because

of it. Specifically, peers successfully identify cognitive skill and but not social skill of their classmates. Referrers frequently choose the same classmates for both skills, at rates much higher than the actual overlap between those who are productive at both cognitive and social skill. For this reason we separately analyzed the three referral strategies: Those made in common for both skills, and those made uniquely for cognitive or social skill. When referring the same classmate across skills, referrers choose classmates with higher grades but not higher skills. This suggests an observable proxy such as academic performance interferes with peer productivity assessments. For unique cognitive skill referrals, both grades and measured cognitive skill are equally good predictors of who got nominated. Referrals made uniquely for social skill could not be predicted by neither academic performance or social skill. These findings reveal a nuanced picture of how peer assessments of productivity can depend on how discernible the skill in question is, and how they can be influenced by the availability of other observable proxies for productivity.

We find limited support for a bias affecting lower social classes. When accounting for peer skills, we find bias only among peers who made unique cognitive skill referrals. This represents about one third of the referrers whose referral decisions are best predicted by cognitive skill and grades of their peers. The incentive treatment mitigates the bias for this subset of referrers, while not changing the referral rates of lower social classes among the rest of the referrers who were not biased. Our findings show peer productivity assessments are robust to salient differences between social classes, and provide evidence that existing biases can be remedied with changes in the incentivization structure.

Our study is organically connected to the literature on referral experiments, from which we draw inspiration in the design. Past experimental work on referrals provides causal evidence that peer productivity assessment via referrals tend to have higher productivity in simple online tasks compared to non-referred workers with similar demographics (Pallais & Sands, 2016). Incentives contingent on referral productivity also tend to improve assessment accuracy in both cognitive skill and job specific technical knowledge (Beaman, Keleher, & Magruder, 2018; Beaman & Magruder, 2012). A notable

102 difference with these studies is that they allow referrers to choose any candidate when as- 102  
103 sessing productivity. We differ by imposing common choice sets for referrals and remove 103  
104 the possibility that success in identifying productive others is due to having different 104  
105 candidate pools to pick from (Montgomery, 1991). We instead explore how sustained in- 105  
106 teraction among peers impacts productivity assessments across skills, and provide causal 106  
107 evidence that peers have skill-dependent screening ability under performance contingent 107  
108 incentives. 108

109 We contribute to the growing body of work on the relevance of noncognitive skills 109  
110 in the labor market. Patience, self-control, conscientiousness, teamwork, and critical 110  
111 thinking are all skills that contribute to positive labor market returns (Heckman & 111  
112 Kautz, 2012; Heckman, Stixrud, & Urzua, 2006; Lindqvist & Vestman, 2011; Weinberger, 112  
113 2014). Among these, interpersonal skills are exceptionally relevant for labor market gains 113  
114 in the last two decades as a complement to cognitive skill (Deming, 2017, 2023). Yet, 114  
115 hiring firms report difficulties in assessing social skills in candidates, and applicants are 115  
116 willing to pay substantial sums to convey social skill feedback to employers (Bassi & 116  
117 Nansamba, 2022). While referrers can assess productivity in cognitive skill and job- 117  
118 specific knowledge (Beaman et al., 2018; Beaman & Magruder, 2012; Pallais & Sands, 118  
119 2016), evidence on whether referrals can help screening for social skills is missing. In a 119  
120 highly educated sample, we find that peers refer productive others in cognitive skill but 120  
121 not social skill. 121

122 Our results also speak to the literature on the diversity considerations in referrals. 122  
123 Homophily<sup>1</sup> in referrals drives correlations among social groups’ employment and wages 123  
124 (Calvo-Armengol & Jackson, 2004; Calvó-Armengol & Jackson, 2007), as individuals are 124  
125 tied more often to others with comparable socioeconomic status (Chetty et al., 2022b). 125  
126 Limited interaction across social classes due to spatial segregation is shown to drive at 126  
127 least some of the differences (Chetty et al., 2022a). In this context, efficiency of diversity 127

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<sup>1</sup>A well-documented empirical consistency in sociology where individuals form ties more often with others who are similar to themselves across observable characteristics (McPherson, Smith-Lovin, & Brashears, 2006; McPherson, Smith-Lovin, & Cook, 2001).

treatments in endogenous networks may be constrained by availability. To counter this, we consider a socially diverse university setting where we imposed networks exogenously, and required participants to refer among classmates. Anticipating differences in referral outcomes for low-SES even when networks across social classes overlap by design, we introduced quota-like incentives as a treatment arm to increase referrals to low-SES peers.<sup>2</sup> Our results provide causal evidence that incentives effectively reduce the social class bias found in our setting.

Section 2 begins with the background and setting in Colombia. In Section 3 we present the design of the experiment, including the skill assessment, referral and guessing tasks. In Section 4 we describe the data and procedures. Section 5 discusses the results of the experiment. Section 6 concludes. The Appendix presents and additional tables and figures as well as the experiment instructions.

## 2 Background and Setting

Our study takes place at UNAB, a medium-sized private university in Bucaramanga, Colombia with approximately 6,000 enrolled students. The university’s student body is remarkably diverse with slightly more than half of the students classified as low-SES. This diversity provides a unique research setting, as Colombian society is highly unequal and generally characterized by limited interaction between social classes, with different socioeconomic groups separated by education and geographic residence.<sup>3</sup> Despite significant financial barriers, many lower middle-class families prioritize university education for their children (Hudson & Library of Congress, 2010, p. 103), with UNAB representing

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<sup>2</sup>We design the treatment incentives in inspiration from the success of gender quotas in the affirmative action literature (e.g., Balafoutas and Sutter (2012); Bertrand, Black, Jensen, and Lleras-Muney (2019); Niederle, Segal, and Vesterlund (2013)).

<sup>3</sup>Colombia has consistently ranked as one of the most unequal countries in Latin America (World Bank, 2024), with the richest decile earning 50 times more than the poorest decile (United Nations, 2023). This economic disparity is reflected by a highly stratified society with significant class inequalities and limited class mobility (Angulo, Gaviria, Pérez, & Azevedo, 2012; García, Rodríguez, Sánchez, & Bedoya, 2015).

149 one of the few environments where sustained inter-class contact occurs naturally. 149

150 In 1994, Colombia introduced a nationwide classification system dividing the popu- 150  
151 lation into 6 strata based on housing characteristics and neighborhood amenities.<sup>4</sup> We 151  
152 use this exogenous cutoff as the measure of social class in our experiment: Students in 152  
153 strata 1 to 3 are categorized as low-SES, and those in strata 4 to 6 as high-SES (see 153  
154 Appendix Figure A.1 for a detailed stratum distribution of our sample). 154

155 We invite all students enrolled in two compulsory courses to participate in our exper- 155  
156 iment. Throughout the term, students meet weekly for three-hour sessions. Attendance 156  
157 is mandatory, ~~but enforcement varies across instructors.~~ Both courses are university- 157  
158 wide graduation requirements which result in large variations in academic programs (see 158  
159 Appendix Table A.3) and socioeconomic backgrounds across the classrooms. This setup 159  
160 provides a unique opportunity for collaborative inter-class contact on equal status, whose 160  
161 positive effects on reducing discrimination are casually documented (Lowe, 2021; Mousa, 161  
162 2020; Rao, 2019). 162

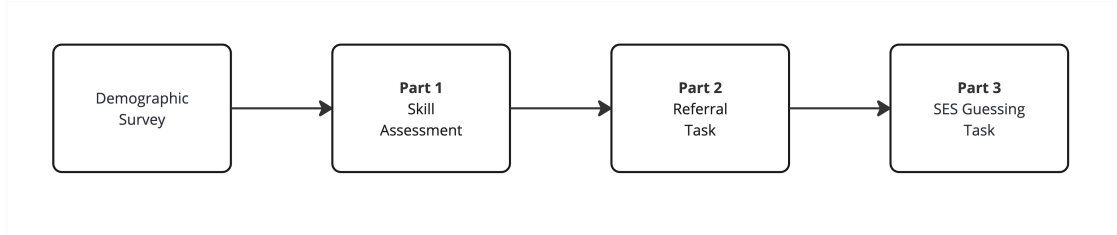
### 163 3 Design 163

164 We designed an experiment to assess the peer screening ability for different skills and to 164  
165 measure biases related to social class. The study design consists of a single experiment 165  
166 with sessions organized at the classroom level (see Figure 1). The instructions are 166  
167 provided in Appendix B. 167

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<sup>4</sup>Initially designed for utility subsidies from higher strata (5 and 6) to support lower strata (1 to 3), it now extends to university fees and social program eligibility. Stratum 4 neither receives subsidies nor pays extra taxes. This stratification system largely aligns with and potentially reinforces existing social class divisions (Guevara S & Shields, 2019; Uribe-Mallarino, 2008).

Figure 1: Experiment Timeline



*Note:* Participants first complete incentivized skill tests, then refer classmates for skills. In the final part, they guess the social class of their peers. This order is implemented in all sessions.

### 3.1 Skill Assessment

To understand the basis for referral decisions, we collect objective measures of cognitive and social skills. These two distinct skills are crucial for the labor market and suitable to assess given classmates interact through the semester. By measuring skills before the referral stage, we eliminated the need for referred students to participate take additional action. Participants perform two incentivized skill tests. They have 5 minutes to complete each test. We provide test-specific instructions and an example item before participants begin. Correctly solved items increase chances to earn a fixed bonus.<sup>5</sup>

We use Raven’s Progressive Matrices to measure cognitive skills (Raven, 1936; Raven, Raven, & Court, 1976). Raven’s test is a well-established measure of fluid intelligence, i.e., an individual’s capacity to reason and solve problems in novel situations independent of past knowledge (Schilbach, Schofield, & Mullainathan, 2016). In this test, participants see series of images where there is a pattern with a piece that has been intentionally removed. They are tasked with choosing the piece that completes the pattern among available options. For each image, there is only one correct answer. We implement an 18-item version featuring increasingly difficult questions, with 6 response options for the

<sup>5</sup>The tests are presented in a randomized order. No performance feedback is provided. Participants see one item at a time and cannot return to previous screens once they start a test. They are not required to answer items and can skip them if they choose to do so. We elicit beliefs about performance after each test.

first 9 items and 8 thereafter.

We measure social skills with the Multiracial Reading the Mind in the Eyes Test (MRMET) from Kim et al. (2022).<sup>6</sup> The test is an established measure for the ability to recognize emotions in others, and it has been previously used in economic experiments (van Leeuwen et al., 2018; Weidmann & Deming, 2021). MRMET consists of photos of human faces portraying different emotions, cropped so that only the eye region is visible. Participants must choose the emotion that best describes the photo from the available answers. For each photo, there is only one correct answer and 4 response options. We administer the first 36 items in MRMET.

### 3.2 Referral Task

After the skill assessment, we create the referral task to screen for high skilled peers. For each skill, participants make incentivized referrals by nominating classmates. We first explain the measured skill accompanied by an example test item. On the following web page, we provide an alphabetically ordered list of all classmates. Participants make three referral choices per skill. They are instructed to exclude themselves from referrals. A classmate may be nominated once per triad. The order in which participants refer for a skill test is randomized. We incentivize referrals with classroom-level performance rankings. The three highest-scoring classmates are designated as the top 3 for a skill. Referrers are eligible for a fixed bonus for referrals among the top 3.<sup>7</sup>

We have two between subject treatments that varies the top 3 selection. In the **Baseline** treatment, the top 3 selection is based solely on performance ranking, regardless of other participant characteristics. The **Quota** treatment modifies the top 3 selection to prioritize low-SES individuals. We reserve the first spot in the top 3 for the highest-scoring low-SES classmate, and assign the remaining two places based on

<sup>6</sup>We choose MRMET because it is a race- and gender-inclusive test suitable for application in non-WEIRD (Western, Educated, Industrial, Rich, Democratic) populations like the one we sample from. The test is based on the original RMET (Baron-Cohen, Wheelwright, Hill, Raste, & Plumb, 2001).

<sup>7</sup>We solve ties among the top 3 randomly. We describe only the top 3 selection mechanism and provide no feedback about the top 3 composition to participants.

performance (see Table 1). This guarantees at least one low-SES participant in the top 3 per skill. Participants are informed about the top 3 selection mechanism before making referral choices (Appendix Figure B.1 provides illustrations explaining the treatments). Assignment to the treatment is at the individual level within each classroom. This allows comparing the effect of the treatment while keeping the referral choice set constant.

Table 1: Places in the Top 3 according to composition rule

|                      | Baseline | Quota |
|----------------------|----------|-------|
| Merit-only           | 3        | 2     |
| Reserved for low-SES | 0        | 1     |

### 3.3 SES Guessing Task

Participants make guesses about the anticipated SES of their classmates. We inform participants that a computer algorithm randomly selects three students belonging to strata 1, 2, or 3. They are tasked with nominating the people they believe the computer could choose at random (Appendix Figure B.2 provides the illustration explaining the task). Participants select three classmates from an alphabetically ordered list containing all their classmates. This task measures the ability to distinguish-SES independent of test performance, as SES identification is relevant to our study.

## 4 Sample, Incentives, and Procedure

We invited 849 UNAB undergraduate students to participate in the experiment. Our final sample consists of 713 individuals who completed the study, resulting in an 83% participation rate. We block randomized participants into treatments balancing gender and social class. Table 2 presents key demographic characteristics and academic performance indicators across treatments (Appendix Table A.1 compares the sample and missing students). The sample is well-balanced between the Baseline and Quota

conditions and we observe no statistically significant differences in any of the reported variables (all  $p$  values  $> 0.1$ ). The sample is characterized by a majority of low-SES students. About one-third of the sample are first-generation college students. The gender distribution is balanced. The mean GPA of 3.95 is consistent across both treatments.

Table 2: Comparison of demographic characteristics and academic outcomes across treatment conditions

|                             | Baseline | Quota | $p$   |
|-----------------------------|----------|-------|-------|
| Low-SES                     | 59%      | 55%   | 0.297 |
| Female                      | 52%      | 47%   | 0.195 |
| Cognitive skill (out of 18) | 10.04    | 10.27 | 0.322 |
| Social skill (out of 36)    | 18.45    | 18.50 | 0.886 |
| GPA (out of 5)              | 3.95     | 3.95  | 0.828 |
| Entry Exam (out of 100)     | 61.85    | 62.17 | 0.638 |
| Age                         | 19.33    | 19.02 | 0.228 |
| First Generation            | 34%      | 37%   | 0.386 |
| Ethnic Minority             | 1%       | 3%    | 0.133 |
| Rural Community             | 30%      | 27%   | 0.308 |
| Has Scholarship             | 1%       | 1%    | 0.916 |
| # Semesters at UNAB         | 3.18     | 3.17  | 0.916 |

*Note:* Low-SES, female, first generation, ethnic minority, rural, and has scholarship represent binary outcomes and  $p$ -values for these variables are from two-sample tests of proportions. For continuous variables (age, GPA, cognitive skill, social skill, entry exam, and semester),  $p$ -values are from two-sample t-tests with equal variances. All reported  $p$ -values are two-tailed.

Participants could earn bonuses worth 100,000 Pesos (about 26 US Dollars) in each part of the experiment. In the first part, we incentivized performance in the skill tests. 20% of participants were eligible for the bonus. We randomly picked one skill test for each eligible participant and drew a number between 1 and 100. The participant received

the bonus if the percentage of correct answers in the selected test exceeded the drawn number. Chances of earning the bonus increased with each correctly solved question by 5.5% ( $=1/18$ ) for the Cognitive Skill test and by 2.78% ( $=1/36$ ) for the Social Skill test.

In the second part, we incentivized referrals among the top 3 performers. 40% of participants were eligible for the bonus. We randomly selected one skill test and one referral for each eligible participant. The participant received the bonus if their referral was among the top 3. In the third part, we incentivized the correct identification of low-SES classmates. 20% of participants in each classroom were eligible for the bonus. We randomly selected one guess for each eligible participant. The participant received the bonus if their guess correctly identified a low-SES classmate. Draws for the bonuses were independent meaning participants could earn multiple bonuses.

Data collection occurred during the last two weeks of April 2024. Our local partner at UNAB, coordinated scheduled classroom visits and recruited research assistants to administer the experiment. Students present in class on the scheduled visit dates participated. Each classroom visit constituted a separate session. There were in total 35 sessions. Participants accessed the Qualtrics-based experiment using their smartphones during these visits. The median time to complete the survey was 20 minutes, with a compensation of \$26 for 117 lottery winners.

## 5 Empirical Analysis

### Can peers screen cognitive and social skills?

Our first goal is understanding whether higher skilled individuals get more referrals. Because every referrer nominates 3 classmates per skill, analyzing only the extensive margin, i.e., whether an individual gets a referral, is not very informative.<sup>8</sup> We consider the percentage share of referrals from individuals in Baseline condition as our dependent variable. This approach combines the intensive and extensive margins and also makes

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<sup>8</sup>Only 86 of the 849 students (10%) never get a referral for either skill.

comparisons across classrooms with different sizes easier.<sup>9</sup> Formally, we define the percentage share of referrals received by individual  $i$  from participants  $j$  in classroom  $c$  and in Baseline condition ( $\forall j \in B_c$ ) for skill  $s \in \{Cognitive, Social\}$  as:

$$y_{ic}^s = \frac{\sum_{j \neq i} r_{ijc}^s}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (1)$$

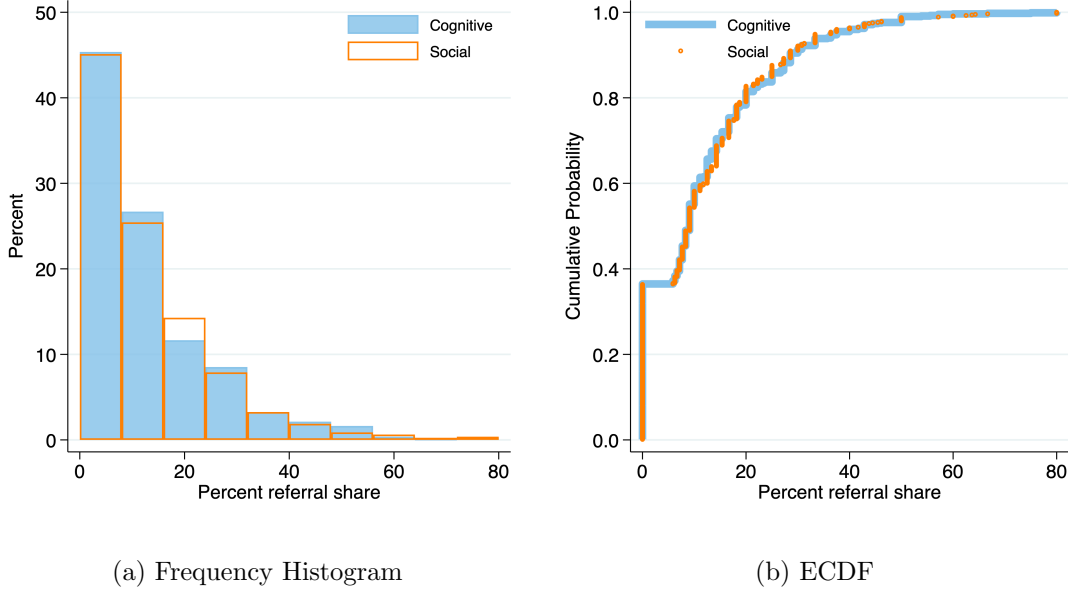
where  $n_c$  represents the number of participants in the Baseline condition in classroom  $c$ . The indicator  $r_{ijc}^s$  takes value 1 if participant  $j$  in the Baseline condition refers individual  $i$  for skill  $s$ , and 0 otherwise, and require both  $i$  and  $j$  to be in the same classroom  $c$ . The denominator  $n_c - \mathbb{1}(i \in B_c)$  accounts for the maximum possible referrals that individual  $i$  could receive. If  $i$  is in the Baseline condition ( $\mathbb{1}(i \in B_c) = 1$ ), we subtract one from  $n_c$  to account for the self-referral restriction.<sup>10</sup> This normalized measure represents the percentage of potential referrals actually received by each individual, adjusting for classroom size and treatment status. By construction,  $y_i^s \in [0, 100]$  for all  $c$ , and we can compare referrals across classrooms of different sizes. Figures 2a and 2b present the distribution of our dependent variable.

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<sup>9</sup>The number of participants in a classroom mechanically drives the number of total referrals that could be received by an individual. By normalizing referrals we focus on differences within classrooms.

<sup>10</sup>33.8 percent of participants in the sample for cognitive and social skills self-referred, while explicitly instructed not to do so. In Appendix Table A.4 compares the outcomes of those who self-refer. Self-referrers are more likely to be low-SES, and have significantly lower cognitive skill (0.2 SD) and GPA (0.25 SD). We rule out the hypothesis that self-referrers nominate themselves strategically. As self-referrers are not informative and add noise to our estimates, we drop these instances from our paired referral-referrer sample in subsequent analyses. Self-referring participants' remaining referral choices are kept in the dataset.

Figure 2: Distribution of referrals by skill in Baseline



*Note:* Figures show the percentage of referrals recieved from participants in the Baseline condition for cognitive and social skills. The left panel shows the frequency histogram and the right panel shows the empirical cumulative distribution function (ECDF). A two-sample Kolmogorov-Smirnov test [show](#) no statistically significant difference between the share of referrals received across the skill distributions ( $D = 0.0363$ ,  $p = 0.668$ ).

Under performance pay in the Baseline condition, higher performing classmates in skill tests should collect more referrals if classmates can succesfully screen skills. Our independent variables are the standardized skill test scores. We estimate referral percentage shares  $y_i^s$ :

$$y_i^s = \beta_0^s + \beta_1^s Skill_i^s + \epsilon_i^s \quad (2)$$

Table 3 illustrates our first findings. The comparison of interest is the point estimates for different skills. In column (1), a one standard deviation increase in cognitive skill causes a 1.2 percentage point increase in the share of referrals received. On a base rate of 13%, this is a modest increase of 9 percent. In column (2), 95% confidence intervals

rule out that a one standard deviation increase in the social skill result in more than a 1 percentage point difference in the share of referrals received. Participants have difficulties screening skills in the Baseline condition, with modest recognition of cognitive and no recognition at all of social skills.



Table 3: Share of referrals received conditional on skill

|              | Cognitive<br>(1)     | Social<br>(2)        |
|--------------|----------------------|----------------------|
| <b>Skill</b> | 1.197**<br>(0.479)   | 0.037<br>(0.510)     |
| Constant     | 12.986***<br>(0.653) | 13.049***<br>(0.521) |
| Observations | 665                  | 665                  |

*Note:* Classroom-level **clustered** standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Dependent variables are the percentage of referrals received relative to all referrals. “Skill” refers to standardized test scores for cognitive and social skills. Sample restricted 665 individuals for whom we have complete administrative and experimental data.

Absence of a clean skill-signal or the lacking the screening ability for skills may have pushed participants to refer classmates using proxies of skills. We define proxies as peer beliefs about strong correlates for skills. A potential proxy for cognitive skill (i.e., “smart students”) would be the “students with good grades” in the classroom, as measured by GPA. The idea that grades signal cognitive skill is a common belief among researchers and practitioners alike. Yet, cognitive skill and grades are far from perfectly correlated (Heckman & Kautz, 2012; Heckman et al., 2006), and screening with such beliefs may not lead to good referrals. Indeed, GPA correlates very weakly with skills in our sample (see Appendix Table A.2). We capture the screening behavior using proxies by including the standardized GPA scores of referrals as an independent variable. We reestimate referral

percentage shares for the Baseline condition:

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$$y_i^s = \beta_0^s + \beta_1^s Skill_i^s + \beta_2^s GPA_i^s + \epsilon_i^s \quad (3)$$

Table 4 illustrates our findings. The comparison of interest is the difference between point estimates for skills and GPA. In column (1), a one standard deviation increase in cognitive skill causes a 0.9 percentage point increase in the share of referrals received when controlling for GPA. On a base rate of 12.8%, this is a comparable increase in magnitude of about 7 percent to our previous estimate in Table 3, and suggests cognitive skills have an independent effect on referrals. However, a one standard deviation increase in GPA causes a substantial 3.9 percentage point increase in the share of referrals received when controlling for cognitive skill. This is an increase of more than four times in terms of magnitude (30 percent) when compared to cognitive skill, and suggestive of the extent to which academic performance is easier to screen among peers in our setting. In column (2), 95% confidence intervals rule out that a one standard deviation increase in the social skill result in more than a 1 percentage point difference in the share of referrals received. This is consistent with our previous estimate confirming participants cannot screen social skills. On the other hand, a one standard deviation increase in GPA causes a substantial 3.4 percentage point increase in the share of referrals received when controlling for social skill. This is a slightly smaller increase of 26 percent in the share of referrals for social skill. For both skills, we find strong evidence that grades act as a proxy for referral decisions. In the next section we expand on the diversity in referral choices to differentiate between referrers using GPA proxy and others.

Table 4: Share of referrals received conditional on skill and academic performance

|              | Cognitive | Social    |
|--------------|-----------|-----------|
|              | (1)       | (2)       |
| Skill        | 0.873*    | -0.278    |
|              | (0.467)   | (0.460)   |
| GPA          | 3.949***  | 3.429***  |
|              | (0.664)   | (0.581)   |
| Constant     | 12.806*** | 12.891*** |
|              | (0.662)   | (0.636)   |
| Observations | 665       | 665       |

*Note:* Classroom-level clustered standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . Dependent variables are the percentage of referrals received relative to all referrals. “Skill” refers to standardized test scores for cognitive and social skills. GPA is standardized to mean zero and unit variance.

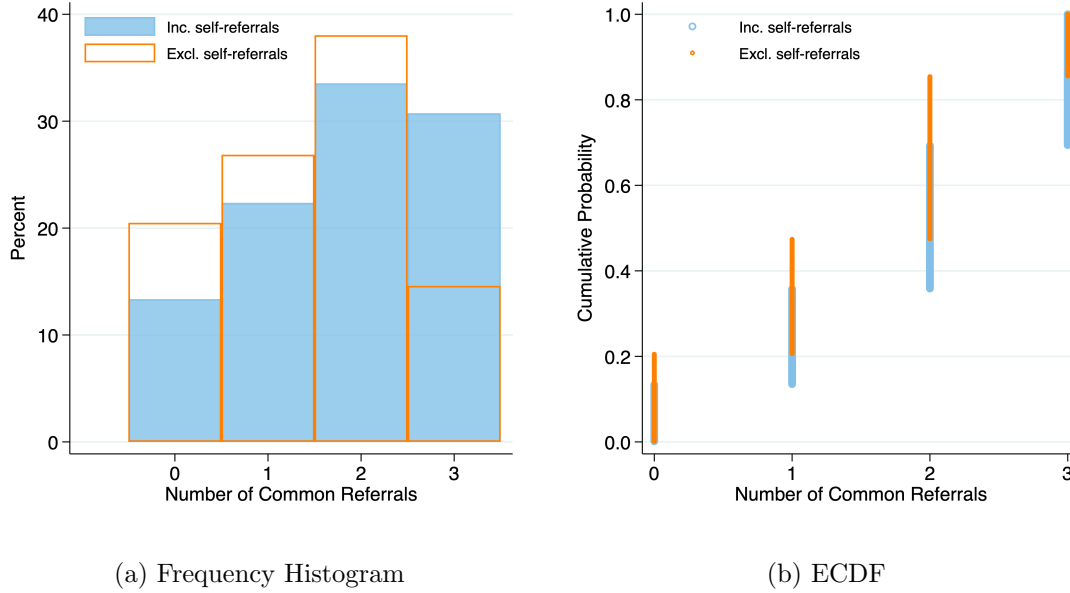
## Types of Referrals

Despite having the opportunity to nominate up to six different classmates across two skills, referral choices are highly concentrated. Only 20% of participants made fully unique referrals, while the median participant nominated two classmates twice, effectively using four of their six referral slots for the same individuals. The concentration of referrals becomes even more pronounced when considering self-referrals which illustrate participants’ actual choices.<sup>11</sup> 30% consistently chose themselves and the same two classmates across skills. Most participants nominated two classmates twice for both skills, and picked themselves or someone else with almost equal probability. We visualize referral choices by plotting the number of common referrals made across skills in Figures

<sup>11</sup>Self-referrals were not valid and are excluded from the remaining analyses.



Figure 3: Distribution of common referrals in Baseline



*Note:* Figures show the distribution of common referrals with and without self-referrals. The first bar (value of 0) indicates the share of participants with 6 unique referrals. The last bar (value of 3) indicates the share of participants with 3 identical referral choices across both skills.

With such a large share common referrals across skills, it is possible that participants believed classmates with a higher score in one skill would also have a higher score in the other. Would such beliefs be accurate? There is only modest ( $\rho = 0.267$ ) correlation between the two skills (see Appendix Table A.2). To understand whether making common referrals is strategic, we turn to the incentives. Participants were incentivized to pick the top 3 performers for each skill to earn a fixed bonus. Looking at the characteristics of top skilled participants in Appendix Table A.6, we find that conditional on being

<sup>12</sup>In Appendix Table A.5 we compare the characteristics of referrers who make unique referrals to those who made at least one common referral. Results suggest minimal differences in GPA, skills, and social class.

among the top 3 for one of the skills, only 1 in 3 participants were in the top 3 for both skills. This suggests making more than 1 common referral across skills would decrease the chances to win the bonus.

A competing explanation for the amount of overlapping referral choices coupled with the notable difficulties in screening skills would be that individuals who refer classmates twice for both skills are worse at screening skills. This implies the underlying heterogeneity in skill identification results in differential referral strategies where participants with a good signal for a skill choose to refer classmates only once for that skill, and those without a good signal use the grades proxy and refer classmates for both skills. We can test both hypotheses in our data: If “twice” referrers -defined as those who refer an individual for both skills- are better at screening at least one of the skills, this gives more credence to beliefs about correlated skills. On the other hand, if “twice” referrers are worse in skill identification compared to single-referrers and use GPA proxy for referrals, we can infer that they have no additional information about skills.

We compare the outcomes of participants who receive referrals twice from their classmates to those who receive referrals only once per skill. Formally, let indicator  $r_{ijc}^{twice}$  take value 1 if individual  $j$  referred individual  $i$  for both skills. The percentage share of referrals received by individual  $i$  from participants in classroom  $c$  and in Baseline condition ( $\forall j \in B_c$ ) is:

$$y_{ic}^{twice} = \frac{\sum_{j \neq i} r_{ijc}^{twice}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (4)$$

where  $n_c$  represents the number of participants in the Baseline condition in classroom  $c$ . The indicator  $r_{ijc}^{twice}$  takes value 0 if participant  $j$  in the Baseline condition does not refer individual  $i$  for both skills. The denominator  $n_c - \mathbb{1}(i \in B_c)$  accounts for the maximum possible “twice” referrals that individual  $i$  could receive as before. Similarly, let  $r_{ijc}^{s,single}$  take value 1 if individual  $j$  referred individual  $i$  only for skill  $s$ . The percentage share of “single” referrals received by individual  $i$  from participants in classroom  $c$  and in Baseline condition ( $\forall j \in B_c$ ) for skill  $s$  is:

$$y_{ic}^{s,single} = \frac{\sum_{j \neq i} r_{ijc}^{s,single}}{n_c - \mathbb{1}(i \in B_c)} \times 100 \quad (5)$$

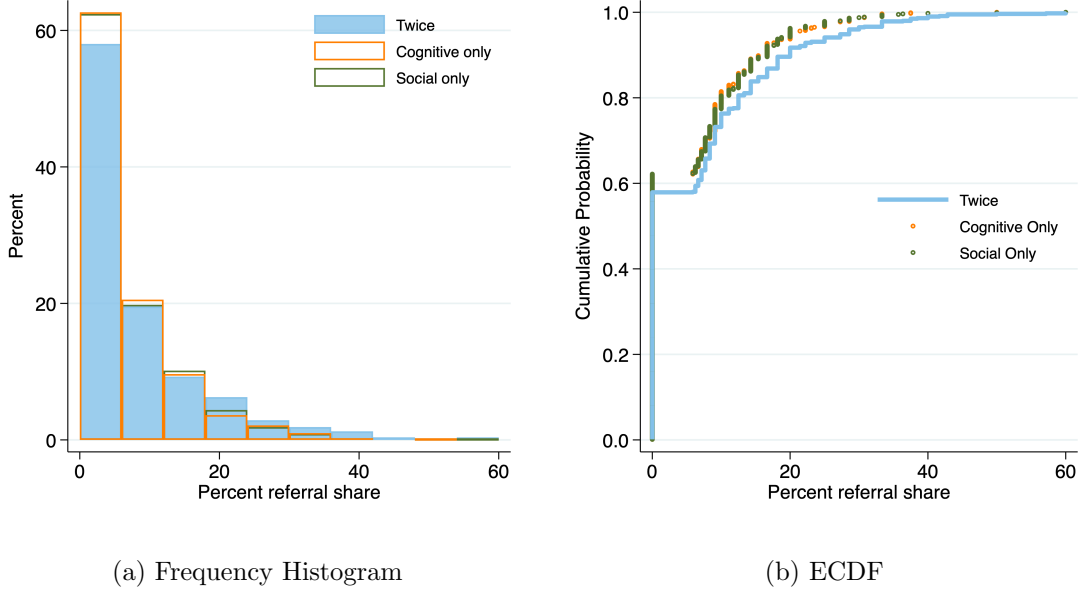
and it follows that for any  $s$ , percentage share of “single” and “twice” referrals received by individual  $i$  from participants in classroom  $c$  and in Baseline condition ( $\forall j \in B_c$ ) must add up to the total share of referrals received:

$$y_{ic}^s = y_{ic}^{s,single} + y_{ic}^{twice} \quad (6)$$

In our data, 48% of all referrals fall under  $y_{ic}^{twice}$ . This is equivalent to saying half of cognitive and social skill referrals were made in common. Figures 4a and 4b present the distribution of the three referral types.



Figure 4: Distribution of referral types in Baseline



*Note:* Figures show the percentage of referrals recieved from participants in the Baseline condition depending on the referral type. The left panel shows the frequency histogram and the right panel shows the empirical cumulative distribution function (ECDF). Two-sample Kolmogorov-Smirnov tests show no statistically significant differences between the share of referrals received between cognitive only and social only referral distributions ( $D = 0.0125$ ,  $p = 1.000$ ) as well as “twice” referrals ( $D = 0.0602$ ,  $p = 0.111$  for cognitive and  $D = 0.0551$ ,  $p = 0.177$  for social).

We regress Equation 3 for our three new dependent variables and report our findings in Table 5. The comparison of interest is the skill and GPA estimates across columns. In column (1), we find that a one standard deviation increase in GPA causes a 3.2 percentage point increase in the share of “twice” referrals received when controlling for skills. This is a staggering 43 percent increase on a base rate of 7.4%. 95% confidence intervals rule out that a one standard deviation increase in either skill result in more than a 1 percentage point difference in the share of “twice” referrals received. Participants who nominate the same individuals for both skills make referrals based on academic performance and cannot observe classmates with higher cognitive or social skills.

For participants who receive only cognitive skill referrals, in column (2), we find that a one standard deviation increase in GPA causes a 0.8 percentage point increase in the share of referrals when controlling for cognitive skill. A one standard deviation increase in cognitive skill causes a comparable 1 percentage point increase in referrals when controlling for GPA. These are respectively 15 and 19 percent increases in the share of referrals received, and suggest participants are able to screen higher skilled peers when referring only for cognitive skill. The lower base rate of 5.4% compared to 7.4% in column (1) suggest less than half of referrals came from “single” referrals. GPA estimate is three times smaller in magnitude compared to column (1), and suggests a smaller weight put in the grades proxy. Nevertheless, the comparable magnitudes of GPA and cognitive skill point estimates still suggest participants refer peers with higher grades much more often than the correlation between the two supported by the data ( $\rho = 0.085$ ). There is heterogeneity in skill identification even for those who refer a peer once for cognitive skill.

For participants who receive social skill referrals only, in column (3), 95% confidence intervals rule out that a one standard deviation increase in social skill or GPA result in more than a 1 percentage point difference in the share referrals received. These further support our previous finding that peers cannot screen social skills in our sample, and attempts to screen social skills by ceasing the use of the GPA proxy seems to fail. Taken together, there are three distinct referral strategies we find in our sample. A majority of individuals cannot screen skills, refer instead using the GPA proxy, and nominate the same individuals for both skills. Those who refer an individual only for cognitive skill can identify this skill, and drive the entirety of the results in terms of peer skill recognition. Still, they confound the cognitive skill signal with academic performance, and put similar weights on the two. Finally, those who refer an individual only for social skill can neither screen social skill or use the GPA proxy. In the subsequent section, we analyze referrals from the perspective of social class while accounting for the referral strategies described in this part.

Table 5: Share of “twice” versus “single” referrals received conditional on skills and academic performance

|                 | Twice<br>(1)        | Cognitive <b>Only</b><br>(2) | Social <b>Only</b><br>(3) |
|-----------------|---------------------|------------------------------|---------------------------|
| GPA             | 3.172***<br>(0.464) | 0.801**<br>(0.391)           | 0.260<br>(0.334)          |
| Cognitive Skill | -0.042<br>(0.416)   | 1.006***<br>(0.270)          |                           |
| Social Skill    | -0.353<br>(0.304)   |                              | 0.086<br>(0.381)          |
| Constant        | 7.407***<br>(0.524) | 5.400***<br>(0.386)          | 5.485***<br>(0.376)       |
| Observations    | 665                 | 665                          | 665                       |

*Note:* Classroom-level clustered standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Independent variables are the respective standardized test scores for skills and GPA.

## Social class bias across referral types

Based on the referral types from the previous section, we document the existence of a social class bias in referrals when controlling for skills and academic performance at Baseline. Our dependent variables are the percentage shares of referrals received at Baseline as defined in Equation 6, and we include a social class dummy for the participant receiving the referrals. We estimate for our three dependent variables:

$$y_i^s = \beta_0^s + \beta_1^s GPA_i + \beta_2^s Skill_i^s + \beta_3^s SES_i + \epsilon_i^s \quad (7)$$

Table 6 summarizes our findings. The comparison of interest is the SES estimates for the three referral strategies. In column (1), controlling for skills and GPA, the point

411 estimate for low-SES is not statistically significant. Skill and GPA estimates are robust 411  
 412 to the inclusion of this variable and remain close to those in Table 5. For participants 412  
 413 who receive cognitive skill referrals only in column (2), we find that being low-SES causes 413  
 414 a 2 percentage points decrease in the share of referrals when controlling for cognitive 414  
 415 skill and GPA. This is a substantial 31 percent difference in the share of referrals re- 415  
 416 ceived, confirming participants are biased against low-SES peers when referring only for 416  
 417 cognitive skill. Skill and GPA estimates are robust to the inclusion of this variable, but 417  
 418 the decrease in the cognitive skill point estimate from 1.006 in Table 5 to 0.869 with the 418  
 419 inclusion of SES dummy suggest the two are confounders. This is supported by the data, 419  
 420 where low-SES underperform in cognitive skill (see Appendix Figure A.3a). GPA and 420  
 421 low-SES are not confounders and there are no differences across social classes in terms 421  
 422 of GPA (see Appendix Figure A.2). For participants who receive social skill referrals 422  
 423 only, in column (3), the point estimate for low-SES is not statistically significant. GPA 423  
 424 estimate remains as before. The decrease in the social skill point estimate with the inclu- 424  
 425 sion of SES dummy suggests the two are confounders, and this is also supported by the 425  
 426 data: Low-SES underperform also in social skill (see Appendix Figure A.3b) consistent 426  
 427 with Falk, Kosse, Pinger, Schildberg-Hörisch, and Deckers (2021). Taken together, we 427  
 428 document a low-SES bias for cognitive skill referrals only. In the next section we explore 428  
 429 whether the social class bias survives when we consider referrals together, and the effects 429  
 430 of the Quota treatment on low-SES referrals. 430

Table 6: Share of “twice” versus “single” referrals received conditional on skills, academic performance, and social class

|                 | Twice<br>(1)        | Cognitive Only<br>(2) | Social Only<br>(3)  |
|-----------------|---------------------|-----------------------|---------------------|
| GPA             | 3.170***<br>(0.462) | 0.797**<br>(0.386)    | 0.260<br>(0.334)    |
| Cognitive Skill | 0.000<br>(0.411)    | 0.869***<br>(0.261)   |                     |
| Social Skill    | -0.306<br>(0.315)   |                       | 0.047<br>(0.372)    |
| Low-SES         | 0.799<br>(0.939)    | -2.017***<br>(0.711)  | -0.549<br>(0.610)   |
| Constant        | 6.948***<br>(0.742) | 6.558***<br>(0.656)   | 5.800***<br>(0.558) |
| Observations    | 665                 | 665                   | 665                 |

*Note:* Classroom-level clustered standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Independent variables are the respective standardized test scores for skills and GPA.

## Social class bias and the Quota treatment

In the following empirical specification, we document whether there is a social class bias when referrals are grouped, and whether the Quota treatment causes referral shares of low-SES participants to change when controlling for skills and academic performance. We hypothesize that the Quota treatment should increase referrals to low-SES classmates because of the additional incentive to refer low-SES. The dependent variable is the percentage share of referrals received as defined for the Baseline treatment in Equation

1, now extended to the referrals from the Quota treatment. It is trivial to see  $y_{ic}^s$  can also be calculated for the Quota treatment as participants in every classroom are randomized into either treatment. Now, every participant is observed twice in the data, and we add a treatment dummy to indicate whether the referrals came from participants in the Baseline or the Quota treatment. We also add a social class dummy for the participant receiving the referrals to our specification. We estimate:

$$y_i^s = \beta_0^s + \beta_1^s Quota_i + \beta_2^s SES_i + \beta_3^s (Quota_i \times SES_i) + \beta_4^s Skill_i^s + \beta_5^s GPA_i + \epsilon_i^s \quad (8)$$

Table 7 illustrates our findings. Our comparison of interest is the effect of the Quota treatment on low-SES peers. In column (1) for cognitive skill, we find that being low-SES decreases the share of referrals recieved by about 1.2 percentage points when controlling for the skill and academic performance in the Baseline condition. This goes down to about 1.5 percentage points in the Quota condition. These differences are not statistically significant, but their sign suggests a relatively large bias against low-SES classmates: A one standard deviation increase in cognitive skill has less than half this magnitude. This finding is coherent with the bias against low-SES classmates found in the previous section, but suggests it is not large enough to not carry over to all cognitive skill referrals considered together. In column (2) for social skill, we find that being low-SES has no statistically significant effect on the share of referrals recieved when controlling for the skill and academic performance. The sign for the point estimate suggest low-SES get slightly more referrals in the Baseline condition, and the Quota reduces this share. The lack of statistical significance in the SES main effect and interaction term suggests considerable noise in both estimates.



Table 7: Share of referrals received by **referrer** treatment, controlling for skill, academic performance, and social class

|                        | Cognitive<br>(1)     | Social<br>(2)        |
|------------------------|----------------------|----------------------|
| Quota                  | -0.073<br>(0.755)    | 0.299<br>(0.716)     |
| Low-SES                | -1.230<br>(1.079)    | 0.364<br>(1.282)     |
| Quota $\times$ Low-SES | -0.167<br>(1.117)    | -0.835<br>(1.181)    |
| Skill                  | 0.594<br>(0.448)     | 0.201<br>(0.426)     |
| GPA                    | 3.184***<br>(0.517)  | 2.819***<br>(0.493)  |
| Constant               | 13.551***<br>(0.986) | 12.706***<br>(0.952) |
| Observations           | 1,330                | 1,330                |

*Note:* Classroom-level clustered standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variables are the percentage share of referrals received for cognitive and social skills. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant’s socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA.

459 **Effects of the Quota treatment across referral types** 459

460 Effects of the social class bias gets diluted across different referral types. A large propor- 460  
461 tion of participants -“twice” referrers- who struggle with skill identification and screen 461  
462 for skills using the academic performance proxy. But there are no SES differences for 462

GPA in our sample. When referrals are made with academic performance in mind, it seems reasonable not to observe a negative bias against low-SES. Then what about skills, knowing that high-SES score higher in both measures? We observe a bias in undersampling from equally well performing low-SES only for “single” cognitive skill referrals, where referrers have a cleaner signal about the skill compared to “single” social skill referrals. We expect the Quota treatment be effective in increasing referrals for low-SES only in a scenario where the skill can be screened, and turn toward our classification of different referral strategies to test this hypothesis. To get clearer estimates for the effects of the Quota treatment on low-SES referrals, we re-estimate the shares of “twice” and “single” referrals. Following the same logic in the section before, we observe every participant twice in each specification, and add a treatment dummy to indicate whether the referrals came from referrers in the Baseline or the Quota treatment. We keep the social class dummy and regress Equation 8 for the three dependent variables.

Table 8 illustrates our findings. The comparison of interest is the SES of the participant receiving the referrals and the effect of the Quota treatment across “twice” and “single” referral types. In column (1), for participants who refer the same peers twice using the academic performance proxy, we find no statistically significant effect of participant SES or the Quota on the referrals share when controlling for skills and academic performance. The signs on the point estimates suggest that directionally low-SES get more referrals in the Baseline treatment and that the Quota reduces this share. For participants who make only cognitive skill referrals, in column (2), we find that being low-SES in the Baseline treatment reduces the percentage share of referrals received by 2.1 percentage points when controlling for the skill and academic performance. This is a very large effect size which translates to a decrease in referral share by 32 percent on a base rate of 6.6%, and is identical to the one found in Table 6. In turn, the Quota treatment increases referrals to low-SES by 1.42 percentage points when controlling for the skill and academic performance. This is also a large effect size that results in an increase in low-SES referral share by 22 percent. For participants who make only social skill referrals, in column (2), we find no statistically significant effect of participant SES

492 or the Quota on the referrals share when controlling for the skill and academic perfor- 492  
493 mance. These are in accordance with our previous findings that social skills cannot be 493  
494 identified and provide no reason to warrant social class bias. Taken together, we find 494  
495 support for our hypothesis that there is a bias against low-SES peers for the skill that is 495  
496 well-identified by peers, and in which low-SES underperform. We find no evidence of a 496  
497 bias when referrals are made based on academic performance where both social classes 497  
498 perform equally well. Reassuringly, we find the bias in cognitive skill referrals is at least 498  
499 partially alleviated by the Quota treatment. Because there is remarkable heterogeneity 499  
500 in the ability to detect SES for both social classes (see Appendix Figure A.4), we find 500  
501 this increase in low-SES referrals satisfying for our sample. 501

Table 8: Share of “twice” and “single” referrals received by referrer treatment, controlling for skill, academic performance, and social class

|                        | Twice<br>(1)        | Cognitive Only<br>(2) | Social Only<br>(3)  |
|------------------------|---------------------|-----------------------|---------------------|
| Quota                  | 0.436<br>(0.817)    | -0.509<br>(0.598)     | -0.136<br>(0.523)   |
| Low-SES                | 0.857<br>(0.920)    | -2.074***<br>(0.722)  | -0.510<br>(0.613)   |
| Quota $\times$ Low-SES | -1.584<br>(1.159)   | 1.417**<br>(0.656)    | 0.750<br>(0.717)    |
| Cognitive Skill        | -0.079<br>(0.374)   | 0.658***<br>(0.201)   |                     |
| Social Skill           | 0.062<br>(0.283)    |                       | 0.158<br>(0.312)    |
| GPA                    | 2.322***<br>(0.330) | 0.858***<br>(0.312)   | 0.502*<br>(0.278)   |
| Constant               | 6.952***<br>(0.727) | 6.591***<br>(0.654)   | 5.765***<br>(0.562) |
| Observations           | 1,330               | 1,330                 | 1,330               |

*Note:* Standard errors in parentheses. \*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ . The dependent variable in column (1) is the percentage share of “twice” referrals received from the same referrer, in column (2) “single” referral share for cognitive skill, and in column (3) for social skill. Quota is a dummy for the referrals received from classmates in the Quota treatment. Low-SES is a dummy for participant’s socioeconomic status. Remaining independent variables are the respective standardized test scores for skills and GPA.

## 6 Conclusion

We find the effects of referrals on skill screening in a diverse university setting with peer interaction. When referral choice sets are fixed and social transfer channels are closed, students in our sample successfully refer higher cognitive skilled peers. Building on past experiments where referrals contain information about productivity (Beaman et al., 2018; Beaman & Magruder, 2012; Pallais & Sands, 2016), our results present a nuanced picture in the efficiency of referrals for screening ability. We find that the skill signal is often confounded with the signal of a proxy, and the magnitude of this misidentification is very large. Because the academic performance proxy proved not accurate, referrals using the proxy resulted in poor referral skill performance overall. The nature of the productivity information conveyed via referrals is very much dependent on the underlying beliefs about the correlation between the proxy and the skill.

Most individuals referred directly using the proxy instead of focusing on the skills. For referrals to be efficient in screening for the skill, referrers may first need to believe they have a clean skill signal. Forcing referrals when skill signal is not clear may have resulted in the proxy strategy taking over the majority of referrals. Making a referral optional as in Beaman et al. (2018); Beaman and Magruder (2012) in one treatment arm could have helped testing this claim, and we leave this for future research. The fact that we observe the proxy strategy proves performance contingent incentives are the right way to elicit referrals in experimental settings.

The evidence for a social class bias is intertwined with the different referral strategies. Low-SES students get a smaller share of referrals only for “single” referrals for cognitive skill, and not for other types of referrals when controlling for productivity. The interpretation we offer is grounded in data and specific to our context. Referrals made with the academic proxy successfully identify high GPA peers. In our university setting, peers know who in their classroom are good students, and social classes are equally good in terms of GPA. This creates little room for bias against low-SES students. In contrast, low-SES students have significantly lower scores in both skill measures. The fact that

we observe a bias only in cognitive skill suggests underperforming low-SES students may be overrepresented, in line with stereotype conceptualization by [Bordalo, Coffman, Gennaioli, and Shleifer \(2016\)](#), but only in the skill which can be effectively screened for.

Our findings contribute to understanding referral mechanisms by showing that even in settings where networks are fixed and social transfers eliminated, referrals remain an imperfect tool for screening worker skills. The effectiveness of referrals depends critically on referrers' ability to observe and accurately assess specific skills, suggesting important implications for how firms structure their referral programs and highlighting the need for complementary screening mechanisms.

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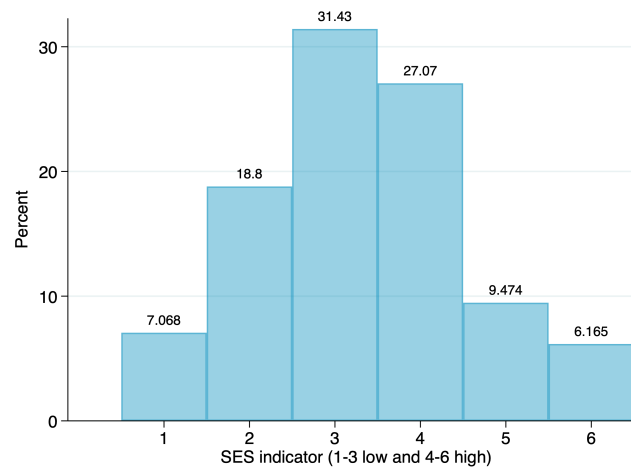
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## A Additional Figures and Tables

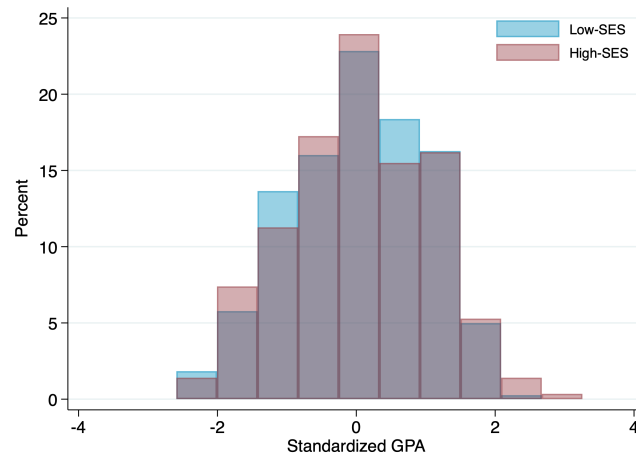
### A.1 Additional Figures

Figure A.1: Stratum distribution of the sample



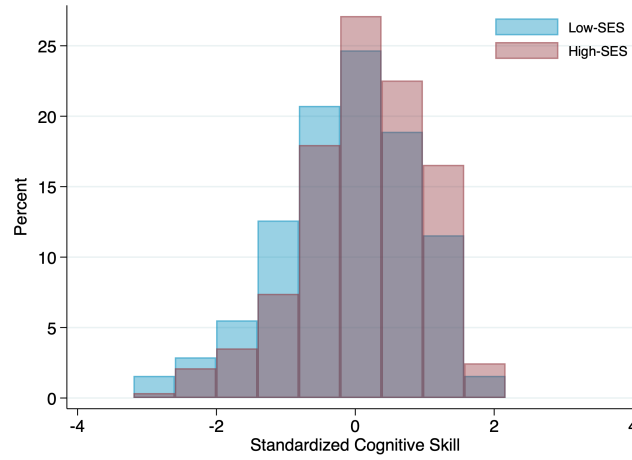
*Note:* Figure shows the distribution of strata in the sample of students that participated in the study.

Figure A.2: GPA by SES

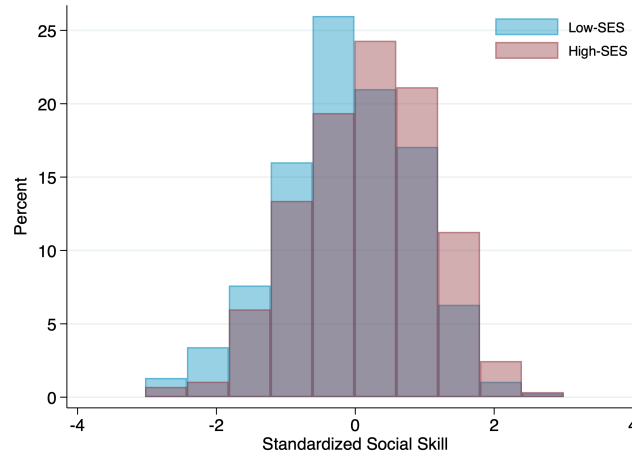


*Note:* Figure shows the distribution of GPA across SES. There are no significant differences in the mean standardized GPA scores between high-SES and low-SES participants ( $t$  test  $p = 0.695$ ).

(a) Cognitive Skill by SES

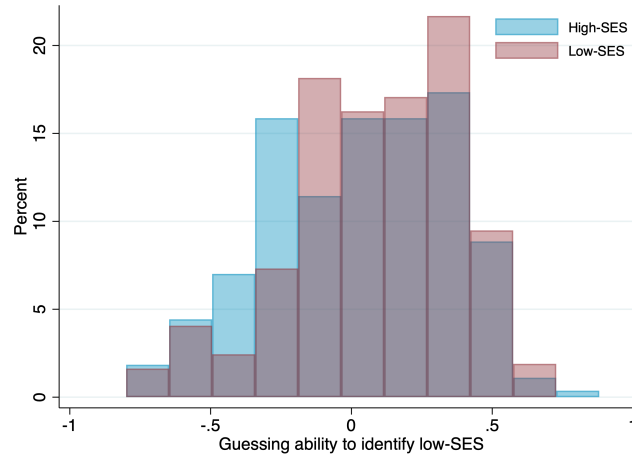


(b) Social Skill by SES



*Note:* Figures show the respective distribution of cognitive and social skills across SES. High-SES outperform Low-SES in both skills ( $t$  tests have  $p$  values  $< 0.001$ ). We can visually verify that larger share of high-SES in quantiles above median for both skills.

Figure A.4: Distribution of guessing ability across SES



*Note:* Figure shows the distribution of the guessing ability across SES. We calculate the guessing ability as the share of succesful low-SES guesses minus the expected probability of randomly drawing low-SES in class  $c$ . A score of 0 indicates an accuracy as good as random draws, below 0 drawing worse than chance, and above 0 better than chance. There are significant differences in the mean guessing ability between high-SES ( $M = 0.022$ ,  $SD = 0.325$ ,  $n = 271$ ) and low-SES participants ( $M = 0.093$ ,  $SD = 0.302$ ,  $n = 369$ ),  $t(638) = -2.85$ ,  $p = 0.005$ ,  $d = 0.226$ . Low-SES participants have higher guessing ability compared to their high-SES counterparts, with a mean difference of 7 percentage points.

Table A.1: Comparison of Variables Between the Sample and Missing Students

|                              | Sample | Missing | <i>p</i> |
|------------------------------|--------|---------|----------|
| Referral share (both skills) | 0.127  | 0.043   | 0.000    |
| GPA (standardized)           | 0.044  | -0.273  | 0.001    |
| Entry Exam (standardized)    | 0.028  | -0.168  | 0.046    |
| # Semesters at UNAB          | 3.171  | 3.188   | 0.884    |
| Age                          | 19.182 | 20.287  | 0.001    |
| Female                       | 49.8%  | 48.5%   | 0.788    |
| Ethnic Minority              | 2.1%   | 4.4%    | 0.114    |
| Rural Community              | 28.8%  | 31.6%   | 0.501    |
| Has Scholarship              | 0.8%   | 0.7%    | 0.899    |

*Note:* Values for female, ethnic minority, rural community, and scholarship represent percentages. All other variables represent means. *p*-values for gender, ethnic, rural, and scholarship are from two-sample tests of proportions. For all other variables, *p*-values are from two-sample t-tests with equal variances. All tests compare the sample and missing students. All reported *p*-values are two-tailed.

Table A.2: Correlation between GPA and skills

|           | GPA    | Cognitive | Social |
|-----------|--------|-----------|--------|
| GPA       | 1.000  |           |        |
| Cognitive | 0.085  | 1.000     |        |
| Social    | 0.093* | 0.267*    | 1.000  |

*Note:* Pairwise correlation between GPA and skills. \* indicates significance at the 5% level with Bonferroni correction. Sample is restricted to 655 participants with complete administrative and experimental data.

Table A.3: Between-Classroom Variation in Academic Programs

| Statistic                 | Most common program share |
|---------------------------|---------------------------|
| Mean                      | 0.424                     |
| Standard Deviation        | 0.216                     |
| 10th percentile           | 0.174                     |
| 25th percentile           | 0.292                     |
| Median                    | 0.345                     |
| 75th percentile           | 0.533                     |
| 90th percentile           | 0.696                     |
| # classrooms with share 1 | 3                         |
| Most diverse classroom    | 0.154                     |
| # classrooms              | 35                        |

*Note:* Table shows the distribution of academic programs across classrooms, measured by the share of students from the most common program in each classroom. Three classrooms are completely homogeneous (share = 1). In the median classroom, the most common program accounts for 34.5% of students. The most diverse classroom has only 15.4% of students in the same program. Data based on 849 students across 35 classrooms.

Table A.4: Characteristics of self-referrers

|                  | No self-referral | Any self-referral | $\Delta$ | $p$   |
|------------------|------------------|-------------------|----------|-------|
| GPA              | 0.132<br>(1.003) | -0.120<br>(0.966) | 0.252    | 0.002 |
| Cognitive Skills | 0.087<br>(0.988) | -0.118<br>(1.023) | 0.205    | 0.013 |
| Social Skills    | 0.034<br>(1.003) | -0.038<br>(0.959) | 0.072    | 0.374 |
| Low-SES          | 0.605<br>(0.490) | 0.511<br>(0.501)  | 0.094    | 0.021 |
| N                | 440              | 225               | 665      |       |
| Share (%)        | 66.2             | 33.8              | 100      |       |

*Note:* Table compares standardized scores between participants who self-referred at least once ( $N = 225$ ) and those who did not ( $N = 440$ ). Positive differences indicate higher scores for those who never self-referred.  $p$ -values from two-sided  $t$ -tests (GPA, Cognitive Skill, Social Skill) and proportion test (Low-SES). The results suggest self-referrers have significantly lower cognitive skills and GPA, and are more likely to be low-SES. Standard deviations in parentheses, samples restricted to participants with complete administrative and experimental data.

Table A.5: Characteristics of participants who make overlapping referrals

|                 | Unique referrals  | Overlapping referrals | $\Delta$ | $p$   |
|-----------------|-------------------|-----------------------|----------|-------|
| GPA             | 0.057<br>(0.983)  | 0.045<br>(1.009)      | 0.012    | 0.903 |
| Cognitive Skill | 0.110<br>(1.005)  | 0.024<br>(0.979)      | 0.086    | 0.371 |
| Social Skill    | -0.014<br>(0.938) | 0.033<br>(0.981)      | -0.047   | 0.621 |
| Low-SES         | 0.530<br>(0.501)  | 0.597<br>(0.491)      | -0.067   | 0.164 |
| N               | 132               | 512                   | 644      |       |
| Share (%)       | 20.5              | 79.5                  | 100      |       |

*Note:* Table compares characteristics between participants who made at least one overlapping referral ( $N = 512$ ) to those who did not ( $N = 132$ , 20.5%). Overlapping referrals indicate cases where a participant referred the same classmate once for cognitive or social skills. Positive differences indicate higher scores for those who made no overlapping referrals. The results suggest minimal differences across all variables.  $p$ -values from two-sided t-tests (GPA, Cognitive Skill, Social Skill) and proportion test (Low-SES). Standard deviations in parentheses, sample restricted to participants with complete administrative and experimental data.

Table A.6: Characteristics of Top Performers and Referrals

|                 | Cognitive        |                  | Social           |                  | Both             |
|-----------------|------------------|------------------|------------------|------------------|------------------|
|                 | Top 3            | Referrals        | Top 3            | Referrals        | Top 3            |
| Cognitive Skill | 1.223<br>(0.419) | 0.112<br>(1.009) | 0.383<br>(0.922) | 0.058<br>(1.015) | 1.201<br>(0.458) |
| Social Skill    | 0.357<br>(0.923) | 0.086<br>(0.996) | 1.340<br>(0.395) | 0.042<br>(1.009) | 1.391<br>(0.453) |
| GPA             | 0.277<br>(0.990) | 0.251<br>(1.021) | 0.264<br>(1.046) | 0.212<br>(1.004) | 0.551<br>(0.897) |
| Low-SES         | 0.457<br>(0.500) | 0.532<br>(0.499) | 0.456<br>(0.500) | 0.555<br>(0.497) | 0.500<br>(0.507) |
| N               | 129              | 1,759            | 114              | 1,775            | 36               |
| Share (%)       | 20.0             | 100              | 17.7             | 100              | 5.6              |

*Note:* Table shows characteristics of students ranked in the top 3 of their classroom and average characteristics of referred students, by skill. Standard deviations in parentheses. Sample restricted to participants with complete administrative and experimental data. All continuous variables are standardized.

## B Experiment

*We include the English version of the instructions used in Qualtrics. Participants saw the Spanish version. Horizontal lines indicate page breaks, and clarifying comments are inside brackets.*

Please enter the password:

[classroom-specific password sent to each participant the day before data collection]

|     |                                                                                               |     |
|-----|-----------------------------------------------------------------------------------------------|-----|
| 663 | <b>Welcome</b>                                                                                | 663 |
| 664 |                                                                                               | 664 |
| 665 | Welcome to this study organized by the Social Bee Lab. You have been invited to               | 665 |
| 666 | participate in a survey where you can make a series of decisions. The study takes ap-         | 666 |
| 667 | proximately 20 minutes to complete. During the study, you should not communicate              | 667 |
| 668 | with any other students. If you have any questions at any time, please raise your hand.       | 668 |
| 669 | One of the assistants will help you privately.                                                | 669 |
| 670 |                                                                                               | 670 |
| 671 | In this study, you can win bonus money depending on your choices. In total, we will           | 671 |
| 672 | draw [classroom-specific number equal to 40% of class size] bonuses of 100.000 pesos          | 672 |
| 673 | among the participants of this classroom. It is also possible for the same person to win      | 673 |
| 674 | more than one voucher. The following screens will detail how the bonus draw will be           | 674 |
| 675 | conducted. The UNAB finance office will make the payment of the vouchers through              | 675 |
| 676 | Nequi.                                                                                        | 676 |
| 677 |                                                                                               | 677 |
| 678 | All your decisions in this survey will be anonymized. Therefore, the answers you provide      | 678 |
| 679 | will not affect your grades in this class or your records at the university. We will use your | 679 |
| 680 | personal information to determine the bonus allocation, but after that, we will remove        | 680 |
| 681 | any data that identifies you.                                                                 | 681 |
| 682 |                                                                                               | 682 |
| 683 | This survey has several parts. Each of these parts has specific instructions. Please read     | 683 |
| 684 | the instructions for each part carefully because they describe how you can earn bonuses.      | 684 |
| 685 | This study has been approved by the [omitted for anonymous review] on the condition           | 685 |
| 686 | that all the information we provide is true and all the bonuses we offer are real.            | 686 |
| 687 | On the next screen, we present you with an informed consent form that you must accept         | 687 |
| 688 | to participate in this study.                                                                 | 688 |
| 689 | <hr/>                                                                                         | 689 |
| 690 |                                                                                               | 690 |
| 691 | <b>Informed Consent</b>                                                                       | 691 |

692 692

693 You have been invited to participate in a study to learn more about how people make 693

694 decisions in common scenarios. 694

695 695

696 This study is conducted by [omitted for anonymous review] and the Social Bee Lab at 696

697 UNAB. The purpose of this study is to broaden our understanding of how people make 697

698 decisions. 698

699 699

700 Participation in this study is voluntary. You may opt-out at any time. No known 700

701 risks are associated with your participation in this project beyond those of everyday life. 701

702 Apart from the monetary bonuses that will be drawn, participation has no direct benefits. 702

703 703

704 The Social Bee Lab is in charge of data collection. Your answers in this study are 704

705 anonymous and will not be shared with anyone. In addition to your answers, UNAB 705

706 will provide the Social Bee Lab with administrative records of your courses and your 706

707 university entrance exam score. Your records, decisions, and your identity will be kept 707

708 strictly confidential. Data about you collected within the scope of the study are used for 708

709 scientific purposes only and are treated as strictly confidential. The Social Bee Lab will 709

710 anonymize your data, and the researcher will analyze it without knowing your identity. 710

711 All data generated will be stored on the researcher's computer. You have the right to 711

712 access your personal data and request its deletion. You can exercise this right by con- 712

713 tacting the researcher. 713

714 714

715 If something is unclear or you have any questions, you can contact [omitted for any- 715

716 mous review]. 716

717 717

718 If you have questions about your rights as a participant, you can contact [omitted for 718

719 anonymous review]. 719

720 720

|     |                                                                                          |     |
|-----|------------------------------------------------------------------------------------------|-----|
| 721 | By continuing to the next screen, you agree to participate in this study.                | 721 |
| 722 |                                                                                          | 722 |
| 723 | <hr/>                                                                                    | 723 |
| 724 |                                                                                          | 724 |
| 725 | Before you start, please answer these four questions.                                    | 725 |
| 726 |                                                                                          | 726 |
| 727 | What is your gender?                                                                     | 727 |
| 728 |                                                                                          | 728 |
| 729 | [Male, Female]                                                                           | 729 |
| 730 |                                                                                          | 730 |
| 731 | What is the socio-economic stratum to which your family belongs?                         | 731 |
| 732 |                                                                                          | 732 |
| 733 | [Stratum 1 to Stratum 6]                                                                 | 733 |
| 734 |                                                                                          | 734 |
| 735 | What is your father's highest acquired level of education?                               | 735 |
| 736 |                                                                                          | 736 |
| 737 | [Primary school, High school, Technical school, Undergraduate, Graduate, Postgradu-      | 737 |
| 738 | ate, Not applicable.]                                                                    | 738 |
| 739 |                                                                                          | 739 |
| 740 | What is your mother's highest acquired level of education?                               | 740 |
| 741 |                                                                                          | 741 |
| 742 | [Primary school, High school, Technical school, Undergraduate, Graduate, Postgradu-      | 742 |
| 743 | ate, Not applicable.]                                                                    | 743 |
| 744 |                                                                                          | 744 |
| 745 | <hr/>                                                                                    | 745 |
| 746 |                                                                                          | 746 |
| 747 | <b>Part 1</b>                                                                            | 747 |
| 748 |                                                                                          | 748 |
| 749 | You will now participate in two quizzes, each lasting five minutes. Please try to answer | 749 |

750 them to the best of your ability. 750

751 751

752 We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 752

753 100.000 pesos in this first part. The steps to allocate the bonuses for Part 1 are explained 753

754 below. 754

755 755

756 \_\_\_\_\_ 756

757 757

758 [classroom-specific illustrations explaining the incentive structure] 758

759 759

760 \_\_\_\_\_ 760

761 761

762 [random assignment to either cognitive or social skills test] 762

763 763

764 **Test - Cognitive Skill** 764

765 765

766 In this test, you will see a series of images. Below is an example of the images you will 766

767 solve. At the top of each image, there is a pattern with a piece that has been removed. 767

768 Your task is to choose which of the six pieces completes the pattern correctly. For each 768

769 image, there is only one correct piece. Look at the following example: 769

770 770

771 First, notice a square in the upper left, the upper right, and the lower left. Also, notice 771

772 that the circle is eliminated when one moves from the upper left to the upper right. 772

773 Finally, the rhombus is eliminated when moving from the upper left to the lower left. 773

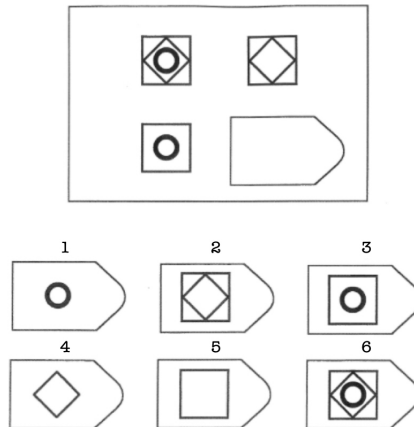
774 Therefore, the correct piece should eliminate the circle and the rhombus, leaving only a 774

775 square. So, the correct answer is piece 5. 775

776 776

777 To give your answer to each image, you must choose the correct option and then continue 777

778 to the next screen. After giving your answer you cannot go back. 778



779

779

780 You will have 5 minutes to complete the test, which consists of 18 images to solve. The  
781 percentage of correct answers will determine your chances of winning one of the 100.000  
782 pesos bonuses if you are chosen for the drawing.

780

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785

786 Are you ready?

786

787

787

788 Your 5 minutes will start as soon as you move to the next screen.

788

789

789

790

790

791

791

792 **Problem 1**

792

793

793

794 [screenshot of Raven's matrix]

794

795

795

796 [After participants submit an answer, a new matrix appears on the screen. The se-  
797 quence of matrices is the same for all participants. Participants cannot return to a

796

797

798 previous screen. Participants do not have to provide answers for all 18 matrices.] 798

799 799

800 \_\_\_\_\_ 800

801 801

802 You have finished the test. You can proceed to the next screen. 802

803 803

804 \_\_\_\_\_ 804

805 805

806 **How did you do on the test?** 806

807 807

808 If we randomly choose 10 participants from this classroom, how many people do you 808

809 think solved fewer correct problems than you? 809

810 810

811 [Slider from 0 to 10] 811

812 812

813 \_\_\_\_\_ 813

814 814

815 **Test - Emotions** 815

816 816

817 In this test, you will see a series of photographs. Below is an example of the pictures you 817

818 will see. In each picture, you will see the eyes of a person. Below the picture, you will 818

819 see four possible emotions that this person is feeling. Your task is to choose which of 819

820 the four emotions correctly describes what the person is feeling. For each picture, there 820

821 is only one emotion. Look at the following example: 821

822 822



823 [Happy, Disappointed, Shocked, Worried] 823

824 824

825 In this case, the correct answer is: Shocked. 825

826 826

827 To give your answer to each picture, you must choose the correct option and then con- 827

828 tinue to the next screen. After giving your answer you will not be able to go back. 828

829 829

830 You will have 5 minutes to complete the test, which consists of 36 photographs to solve. 830

831 The percentage of correct answers you get will determine your chances of winning one 831

832 of the 100.000 pesos bonuses if you are chosen for the drawing. 832

833 833

834 \_\_\_\_\_ 834

835 835

836 Are you ready? 836

837 837

838 Your 5 minutes will start as soon as you move to the next screen. 838

839 839

840 \_\_\_\_\_ 840

841 841

842 **Photograph 1: Choose the word that best describes the photograph** 842

843 843

844 [photo from Multiracial Reading the Mind in the Eyes Test] 844

845 845

846 [After participants submit an answer, a new photo appears on the screen. The sequence 846

847 of photos is the same for all participants. Participants cannot return to a previous screen. 847

848 Participants do not have to provide answers for all 36 photos.] 848

849 849

850 \_\_\_\_\_ 850

851 851

852 You have finished the test. You can proceed to the next screen. 852

853 853

854 

---

 854

855 855

856 **How did you do on the test?** 856

857 857

858 If we randomly choose 10 participants from this classroom, how many people do you 858

859 think solved fewer correct photographs than you? 859

860 860

861 [Slider from 0 to 10] 861

862 862

863 

---

 863

864 864

865 **Part 2** 865

866 866

867 At the beginning of this study, all participants took two tests, one on cognitive ability 867

868 and one on emotions. In this part, we will ask you to recommend the people who in 868

869 your opinion will score the best on each test. 869

870 870

871 You may recommend 3 people per test, but you may not recommend yourself. 871

872 872

873 We will allocate up to [classroom-specific number equal to 40% of class size] bonuses of 873

874 100.000 pesos for Part 2. The steps for allocating bonuses are explained below. 874

875 875

876 

---

 876

877 877

878 [random assignment to either quota or baseline condition] 878

879 879

880 [classroom-specific illustrations explaining the incentive structure depending on assign- 880

881 ment to either baseline or quota conditions]

881

882

882

Figure B.1: Illustrations for the two conditions



883

883

884

884

885 [random assignment to either cognitive or social skills referral task]

885

886

886

## 887 Recommendation - Cognitive Skill

887

888

888

889 All participants took a test to identify the missing pattern in each image, as in the ex-  
890 ample below. This test is used to measure general intelligence.

890

891

891

892 Next, we will present you with a list of the names of all the students in this room. We  
893 will ask you to recommend the three people you think will score the highest on the  
894 general intelligence test.

894

895

895

896 If you are chosen by the computer, each of your recommendations in the top 3 increases  
897 your chances of winning one of the 100.000 pesos bonuses.

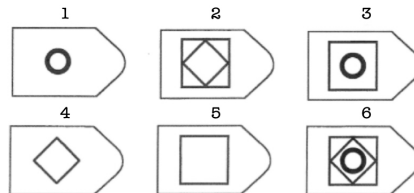
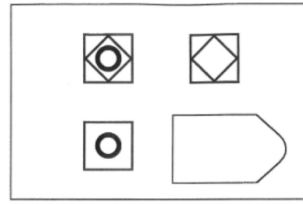
897

898

898

899

899



900  
901 Select the students in this classroom who you consider to have the highest scores on the 901  
902 general intelligence test. (Select 3 students) 902

903  
904 [Classroom-specific list of all classmate names visible on one screen. Participants have 904  
905 to pick 3 classmates to continue. Picking their own name invalidates their choices.] 905

906  
907 \_\_\_\_\_ 907

908  
909 **Recommendation - Emotions** 909

910  
911 All participants took a test where they had to identify the emotion that best described 911  
912 the expression of each image as in the example below. This test is used to measure social 912  
913 skills. 913



915 Next, we will present you with a list of the names of all the students in this room. We 915  
916 will ask you to recommend 3 people you think will score the highest on the social skills 916  
917 test. 917

918 918  
919 If you are chosen by the computer, each of your recommendations in the top 3 increases 919  
920 your chances of winning one of the 100.000 pesos bonuses. 920

921 921

922 \_\_\_\_\_ 922

923 923

924 Select the students in this classroom who you consider to have the highest scores on the 924  
925 social skills test. (Select 3 students) 925

926 926

927 [Classroom-specific list of all the names visible on one screen. Participants have to pick 927  
928 3 classmates to continue. Picking their own name invalidates their choices.] 928

929 929

930 \_\_\_\_\_ 930

931 931

932 **Part 3: Recommendation - Random draw** 932

933 In this part, the computer will randomly choose three students who belong to strata 1, 933  
934 2, or 3. We will ask you to nominate three people you think the computer will choose. 934

935 935

936 We will allocate up to [classroom-specific number equal to 20% of class size] bonuses of 936  
937 100.000 pesos for Part 3. The steps for allocating the bonuses are explained below. 937

938 938

939 \_\_\_\_\_ 939

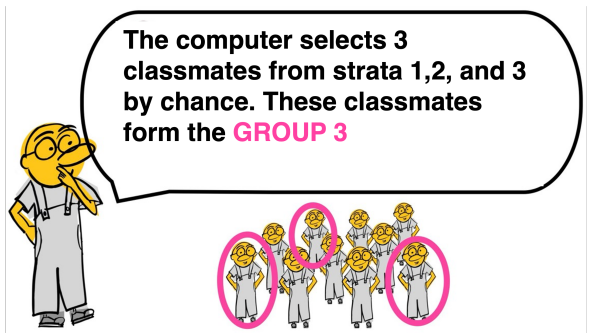
940 940

941 [classroom-specific illustrations explaining the incentive structure] 941

942 942

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Figure B.2: Illustration for the Guessing Task



944 944

945 Select the students in this classroom who belong to strata 1, 2, or 3, who you think will 945

946 be randomly selected by the computer (Select 3 students). 946

947 947

948 [Classroom-specific list of all the names visible on one screen. Participants have to pick 948

949 3 classmates to continue. Picking their own name invalidates their choices.] 949

950 950

951 951

952 952

953 **Part 4** 953

954 Do you want to know your scores on the general intelligence test and the social skills 954

955 test? We can analyze the data and give you a report that explains your strengths in 955

956 these two areas. Also, what do these strengths mean, and how can you leverage them 956

957 for your personal and professional development? 957

958 958

959 If you want to receive your skills report, we need to contact you again. We also want to 959

960 be able to invite you to new studies where you can participate for more bonus money. 960

961 Please indicate if you agree to be contacted again. 961

962 962

963 [I can be contacted for new studies and to send me my report. I can be contacted to 963

964 send my report, but not for new studies. No, I do not want to be contacted again.] 964  
965 \_\_\_\_\_ 965  
966 966  
967 [if participant gives consent to be contacted again] 967  
968 968  
969 Please enter your contact email: 969  
970 970  
971 [student email] 971  
972 972